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ANALYSIS OF NOMINAL GDP OF THE GULF STATES FROM 2000 TO 2025

General Characteristics of the Region

The aggregate nominal GDP of the GCC countries grew from USD 380.65 billion in 2000 to USD 2,456.92 billion in 2025, representing a total increase of 545.5% and a compound annual growth rate (CAGR) of 7.75% [1, 2, 3, 4, 5, 6, 7]. In absolute terms, the increase amounted to USD 2,076.27 billion. Unlike GDP at PPP, the nominal indicator in US dollars directly reflects the region's capacity to generate foreign currency earnings for international investment and imports. As clearly demonstrated by the "Saudi Arabia Share (%)" column in Table 1, the financial weight of Saudi Arabia in the region is not diminishing but, on the contrary, consolidating during periods of high hydrocarbon prices, reaching a historical maximum of 53.63% by 2023.

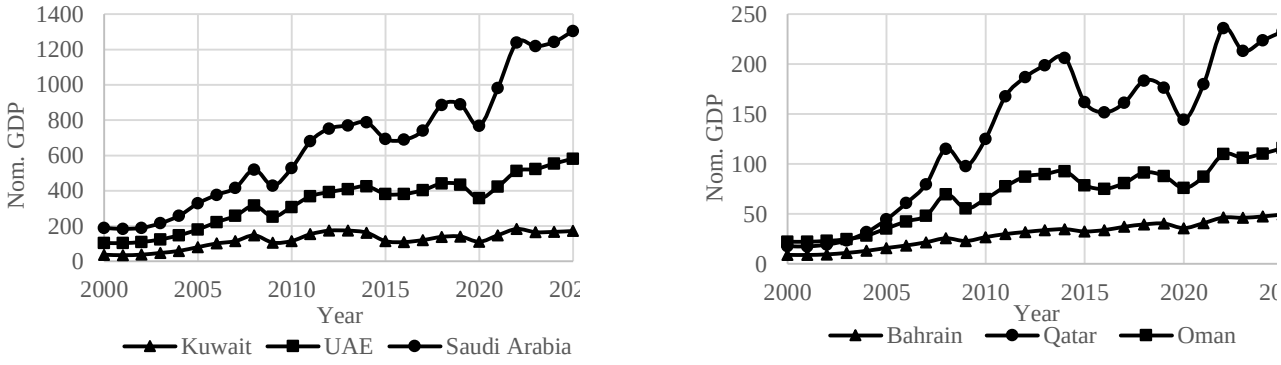
Table 1. Nominal GDP (USD billion)

No.	Year	Bahrain	Qatar	Kuwait	UAE	Oman	Saudi Arabia	SA Share, %
1	2000	9,06	17,76	37,72	104,34	22,26	189,51	49.79
2	2001	8,98	17,54	34,89	103,31	22,21	184,14	49.62
3	2002	9,59	19,36	38,14	109,82	23,07	189,61	48.67
4	2003	11,07	23,53	47,87	124,35	24,73	215,81	48.24
5	2004	13,15	31,73	59,44	147,82	28,38	258,74	47.98

6	2005	15,97	44,53	80,80	180,62	35,51	328,46	47.89
7	2006	18,50	60,88	101,56	222,12	42,58	376,90	45.82
8	2007	21,73	79,71	114,63	257,92	48,30	415,96	44.33
9	2008	25,71	115,27	147,38	315,47	69,80	519,80	43.55
10	2009	22,94	97,80	105,97	253,55	55,45	429,10	44.47
11	2010	26,81	125,12	115,42	307,74	64,99	528,21	45.21
12	2011	29,91	167,78	154,04	368,88	77,50	680,66	46.03
13	2012	31,96	186,83	174,05	392,79	87,41	751,92	46.27
14	2013	33,82	198,73	174,17	409,63	89,94	769,76	45.93
15	2014	34,77	206,22	162,65	424,94	92,70	787,15	46.07
16	2015	32,52	161,74	114,59	381,97	78,71	693,41	47.40
17	2016	33,88	151,73	109,41	381,72	75,13	689,28	47.83
18	2017	37,20	161,10	120,69	403,37	80,86	741,27	48.00
19	2018	39,57	183,33	138,65	440,56	91,51	886,56	49.80
20	2019	40,45	176,37	140,86	433,93	88,06	888,89	50.26
21	2020	35,84	144,41	111,05	357,16	75,91	767,95	51.46
22	2021	40,84	179,73	148,35	422,44	87,32	982,66	52.79
23	2022	46,46	235,71	183,50	511,40	109,85	1239,08	53.27
24	2023	46,19	213,00	165,46	522,62	106,17	1218,58	53.63
25	2024	47,58	223,65	167,11	553,98	110,42	1242,95	52.99
26	2025	49,48	232,60	172,12	581,68	115,94	1305,10	53.12

Note: Data for 2000–2023 are actual values from the World Bank; for 2024–2025, *they are IMF forecast estimates. The "Saudi Arabia Share (%)" column shows the proportion of Saudi Arabia in the aggregate nominal GDP of the six GCC countries* [1, 2, 3, 4, 5, 6, 7].

Figure 1, constructed on the basis of Table 1, illustrates how the nominal GDP values of the GCC countries change over the years.



a *b*

Fig. 1. Dynamics of Nominal GDP of the GCC Countries, USD Billion

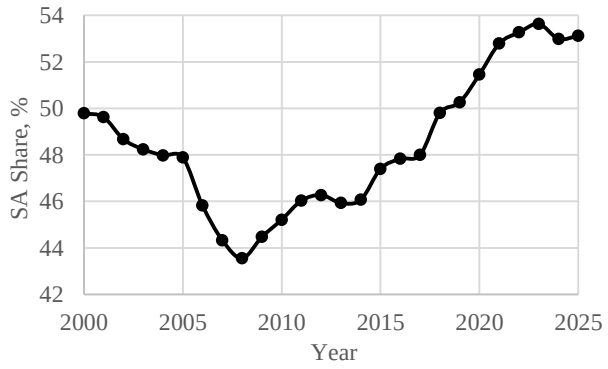


Fig. 2. Share of Saudi Arabia in the Aggregate Nominal GDP of the GCC Countries

Bahrain: Financial and Industrial Hub

Bahrain is the smallest economy in the region. Nominal GDP grew from USD 9.06 billion in 2000 to USD 49.48 billion in 2025 [1] (increase of 446.1%, CAGR 7.02%). Bahrain's share is stable at approximately 2.0% [1, 2, 3, 4, 5, 6, 7]. Bahrain has historically positioned itself as the banking center of the Middle East and also possesses a powerful aluminum industry (Alba). This ensures currency inflows even during periods of declining oil prices [1].

Kuwait: Raw Material Dependence and Stagnation

Kuwait demonstrates the weakest growth rates in nominal GDP. The indicator grew from USD 37.72 billion in 2000 to USD 172.12 billion in 2025 [3], representing an increase of 356.3% and a CAGR of 6.25%—the minimum in the GCC [3]. Kuwait's share declined from 9.9% in 2000 to 7.0% in 2025 (−2.9 percentage points) [1, 2, 3, 4, 5, 6, 7]. Kuwait's economy is critically dependent on crude oil exports, while

diversification is proceeding extremely slowly due to political conflicts between the government and parliament. Kuwait's nominal GDP still cannot confidently rise above the peak values of 2012–2013 (approximately USD 174 billion) [3].

Oman: Modest Reserves and Fiscal Balance

Oman ranks fifth. Nominal GDP grew from USD 22.26 billion in 2000 to USD 115.94 billion in 2025 [5] (increase of 420.8%, CAGR 6.82%). Oman's share declined from 5.8% to 4.7% [1, 2, 3, 4, 5, 6, 7]. Oman has smaller oil and gas reserves than its neighbors, so its nominal dollar revenues are lower. However, the country is actively developing port logistics (Salalah and Duqm ports) and tourism, which helps smooth the effects of oil shocks and maintain moderate growth [5].

Qatar: Gas Superpower

Qatar demonstrates an absolutely phenomenal growth in nominal GDP, unparalleled in the region. GDP grew from USD 17.76 billion in 2000 to USD 232.60 billion in 2025, representing a colossal increase of 1,209.7% and a CAGR of 11.05%—the highest among all GCC countries [2]. Qatar's share in regional nominal GDP more than doubled: from 4.7% in 2000 to 9.5% in 2025 (+4.8 percentage points) [1, 2, 3, 4, 5, 6, 7]. The key driver was liquefied natural gas (LNG) exports, whose contracts have historically been tied to oil prices and denominated in US dollars. The peak of 2022 (USD 235.71 billion) clearly demonstrated how global demand for gas from Europe and Asia instantly converts into Qatar's dollar revenues [2].

Saudi Arabia: Financial Giant and the Effect of Expensive Oil

Saudi Arabia remains the undisputed leader in the region in terms of nominal GDP. In 2000, its economy was valued at USD 189.51 billion, and by 2025, it is projected to reach USD 1,305.10 billion. The total growth over 25 years amounted to 588.7%, with an average annual growth rate of 8.01% [6]. Since the Saudi riyal is rigidly pegged to the US dollar, the country's nominal GDP is effectively a direct function of oil production volumes and global oil prices.

Of particular interest is the dynamics of Saudi Arabia's share in the region's aggregate nominal GDP (see the rightmost column of Table 2.1), which demonstrates

a paradox directly opposite to the trends in real GDP. If in physical terms (real GDP) the kingdom's share declined from 51.7% to 44.7% over 25 years due to Qatar's gas surge and the UAE's diversification, then in dollar equivalent the picture is diametrically different.

In 2000, Saudi Arabia controlled 49.79% of all nominal dollar revenues in the region. During the oil boom of 2006–2008, when oil prices hit historical records and neighbors were actively expanding non-oil exports, Saudi Arabia's share temporarily dropped to a historical minimum of 43.55% (2008). However, starting from 2015, and especially after 2021, the trend reversed. By 2023, Saudi Arabia's share in the GCC nominal GDP reached 53.63%—an absolute maximum for the entire observation period.

This phenomenon is explained by the fact that during periods of high oil prices (above USD 80 per barrel), the colossal absolute volumes of Saudi production generate such a massive currency inflow that it mathematically outweighs any successes of neighbors in diversification. The UAE's economy may be more balanced, and Qatar's may be more technologically advanced in the LNG sector, but Saudi Arabia's "oil cash register" in dollar equivalent remains unreachable.

The decline in 2020 amounted to 13.6% (from USD 888.89 billion in 2019 to USD 767.95 billion in 2020) against the backdrop of the Brent crude price collapse and the pandemic [6]. However, already in 2022, amid the global energy crisis, Saudi Arabia's nominal GDP made a historic leap of 26.1%, for the first time exceeding one trillion dollars (USD 1,239.08 billion) and providing the kingdom with more than half of the entire financial weight of the Persian Gulf [6]. The IMF forecast for 2025 (USD 1,305.10 billion) assumes the continuation of this financial hegemony [7].

United Arab Emirates: Diversification and Regional Hub Status

The UAE ranks second in the region in terms of nominal GDP. In 2000, the economy was valued at USD 104.34 billion, and in 2025—at USD 581.68 billion [4]. Total growth: 457.5%, CAGR: 7.11% [4]. The UAE's share in regional nominal GDP

declined from 27.4% in 2000 to 23.7% in 2025 (−3.7 percentage points) [1, 2, 3, 4, 5, 6, 7]. This is due to the faster growth rates of Saudi Arabia and Qatar. Nevertheless, the UAE possesses a unique structure: in addition to oil, a significant portion of nominal dollar revenues comes from tourism, re-export, logistics, real estate, and financial services. This makes the UAE's economy more resilient to hydrocarbon price declines compared to its neighbors: in the crisis year of 2020, the decline was 17.6%, which is less than in many oil-dependent economies [4].

1.2. Comparative Analysis

Ranking of GCC Countries by Nominal GDP Growth Rates (CAGR, 2000–2025)

Table 2. Ranking of GCC Countries by Nominal GDP Growth Rates

No.	Rank	Country	CAGR, %	Total Growth, %
1	1	Qatar	11.05	1209.7
2	2	Saudi Arabia	8.01	588.7
3	3	UAE	7.11	457.5
4	4	Bahrain	7.02	446.1
5	5	Oman	6.82	420.8
6	6	Kuwait	6.25	356.3

Source: Calculations were made based on data [1, 2, 3, 4, 5, 6, 7].

Changes in the Structure of the Regional Economy

In 2000, the top three (Saudi Arabia—49.8%, UAE—27.4%, Kuwait—9.9%) controlled 87.1% of regional nominal GDP. In 2025, their combined share decreased to 83.8% (Saudi Arabia—53.1%, UAE—23.7%, Kuwait—7.0%); however, Saudi Arabia strengthened its absolute dominance. The main beneficiary was Qatar, whose share nearly doubled (+4.8 percentage points) due to gas monetization [1, 2, 3, 4, 5, 6, 7].

Volatility and Specifics of Nominal GDP

Nominal GDP in US dollars is significantly more volatile than GDP at PPP. This is because it includes not only the physical growth of economies but also US dollar inflation, as well as sharp fluctuations in exchange-traded oil and gas prices. The approximate coefficient of variation of annual growth rates (2001–2024):

- Qatar: ~105.4% (extreme volatility due to LNG prices);
- Kuwait: ~88.2%;
- Saudi Arabia: ~72.5%;
- UAE: ~65.8%;
- Oman: ~58.3%;
- Bahrain: ~49.1% (minimum volatility due to diversification).

Source: Calculations were made based on data [1, 2, 3, 4, 5, 6].

Impact of Oil Shocks

Analysis of the dynamics of nominal GDP of the GCC countries reveals the region's high sensitivity to global economic shocks. The dot-com crisis of 2001 was the first serious test, causing a GDP decline in all countries of the region, with Kuwait suffering the most significantly (−9.6%) due to the high dependence of its economy on oil revenues and weak diversification. The global financial crisis of 2008–2009 led to a collapse of GCC nominal GDP by 19.2% (from USD 1,193.43 billion in 2008 to USD 964.81 billion in 2009) due to the fall in oil prices from USD 140 to USD 40 per barrel [1, 2, 3, 4, 5, 6]. The oil shock of 2014–2016 provoked a decline of 15.6% from the peak value of 2014. The COVID-19 pandemic of 2020 caused a decrease of 15.6% (from USD 1,768.56 billion in 2019 to USD 1,492.32 billion in 2020) [1, 2, 3, 4, 5, 6]. Finally, the correction of 2023, associated with voluntary production cuts under OPEC+ agreements, led to a decline after the record year of 2022, especially in Saudi Arabia (−3.6%).

Recovery After Crises

Unlike GDP at PPP, the recovery of nominal GDP strongly depends on commodity market conditions. After the 2009 shock, the region returned to pre-crisis

levels only in 2011. After the 2020 pandemic, recovery was rapid: already in 2021, aggregate nominal GDP exceeded the 2019 level, and in 2022, it reached a historic maximum of USD 2,326.00 billion thanks to the geopolitical risk premium in energy markets [1, 2, 3, 4, 5, 6].

** In this context, CAGR for nominal GDP has an important feature: unlike GDP at PPP (which smooths out price differences), nominal CAGR in US dollars includes the effect of US dollar inflation over these* 25 *years. That is, part of the* 7.75*% growth is not a physical increase in oil or service production, but the depreciation of the currency itself in which calculations are made* [9].

CONCLUSIONS

1. Nominal vs. Real GDP Paradox of Saudi Arabia: The growth of Saudi Arabia's share in nominal GDP to 53.63% alongside a decline in real GDP to 44.7% indicates that the nominal indicator reflects price conditions and monetary factors rather than physical economic growth.
2. Qatar as the Sole Structural Transformer of the Region: A CAGR of 11.05% and an increase in share from 4.7% to 9.5% demonstrate the creation of a new power center in the GCC through gas monetization.
3. Kuwait is Systemically Losing Ground: The lowest CAGR (6.25%), a decline in share from 9.9% to 7.0%, and the inability to surpass the 2012-2013 peak point to a structural crisis rather than a cyclical downturn.
4. UAE Chose Stability Over Growth: The reduction in share from 27.4% to 23.7% represents a deliberate strategic choice in favor of diversification and resilience against oil shocks.
5. Classic Risk/Return Trade-off in the GCC: Bahrain (volatility 49.1%, CAGR 7.02%) vs. Qatar (volatility 105.4%, CAGR 11.05%) demonstrates the direct relationship between high returns and high risk.
6. Nominal GDP as an Indicator of Geopolitical Power, Not Economic Development: The 588.7% growth is a function of oil production volume and global oil prices, making this indicator a measure of currency power rather than real welfare.

7. The "Oil Cash Register" Effect of Saudi Arabia: During periods of high oil prices (>$80/barrel), Saudi Arabia's absolute production volumes mathematically overshadow any success of neighboring countries in diversification.
8. Nominal GDP CAGR Includes Dollar Inflation: Part of the 7.75% growth represents not a physical increase in production, but the depreciation of the currency used for calculations.
9. Recovery Depends on External Factors: The speed of recovery after crises is determined by commodity market conditions rather than the quality of domestic reforms (2009→2011: 2 years; 2020→2021: 1 year).
10. Structural Shift Towards Multipolarity: The share of the top three (Saudi Arabia, UAE, Kuwait) declined from 87.1% to 83.8%, indicating a relative increase in the weight of Qatar, Oman, and Bahrain.

REFERENSES

1. World Bank. GDP (current US$) – Bahrain // World Bank Open Data. — Washington, D.C., 2026. — URL: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=BH (date of access: 21.06.2026). — Text: electronic.
2. World Bank. GDP (current US$) – Qatar // World Bank Open Data. — Washington, D.C., 2026. — URL: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=QA (date of access: 21.06.2026). — Text: electronic.
3. World Bank. GDP (current US$) – Kuwait // World Bank Open Data. — Washington, D.C., 2026. — URL: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=KW (date of access: 21.06.2026). — Text: electronic.
4. World Bank. GDP (current US$) – United Arab Emirates // World Bank Open Data. — Washington, D.C., 2026. — URL:

https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=AE (date of access: 21.06.2026). — Text: electronic.

5. World Bank. GDP (current US$) – Oman // World Bank Open Data. — Washington, D.C., 2026. — URL: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=OM (date of access: 21.06.2026). — Text: electronic.
6. World Bank. GDP (current US$) – Saudi Arabia // World Bank Open Data. — Washington, D.C., 2026. — URL: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=SA (date of access: 21.06.2026). — Text: electronic.
7. International Monetary Fund. World Economic Outlook Database, April 2026: GDP, current prices (U.S. dollars) // IMF Data. — Washington, D.C., 2026. — URL: https://www.imf.org/external/datamapper/NGDPD@WEO/ (date of access: 21.06.2026). — Text: electronic.
8. Organization of the Petroleum Exporting Countries. Monthly Oil Market Report, June 2022. — Vienna: OPEC, 2022. — URL: https://www.opec.org/opec_web/en/publications/338.htm (date of access: 21.06.2026). — Text: electronic.
9. World Bank. DataBank Methodology: Average annual growth rate // World Bank Open Data. — Washington, D.C., 2026. — URL: https://databank.worldbank.org (date of access: 21.06.2026). — Text: electronic.